

Advantages over proprietary ISA

The RISC-V ecosystem's value lies not just in potentially lower licensing costs compared to traditional ISAs like ARM but in the broader benefits it offers to the design community.

Supplier independence. RISC-V software is largely independent of any hardware supplier, allowing customers to switch providers with minimal impact on their software tools or applications. This flexibility was not possible with proprietary ISAs, where changing ISAs meant a complete software overhaul.

Greater selection. RISC-V International coordinates the standardisation of the ISA but allows developers to customise their own extensions without needing to filter changes back through the association. This contrasts with proprietary ISAs, where a single company controls both the specifications and implementations.

Freedom to innovate. Closed ISAs restrict customisation, unless a customer agrees to an architectural licence with potential cost and constraints. RISC-V, as an open standard, can be obtained free of charge, allowing for innovation without the need for expensive licensing.

Quicker and greater innovation. RISC-V's open nature means that innovations can come from multiple companies, each focusing on different markets and applications. This collective approach can lead to faster and more diverse innovations than a single company could achieve..

Risk-V

This is a computing era. Increased demand for automation, endless availability of connections, and to top it off, artificial intelligence has exploded the limits of what is possible. The expansion of IoT sensors, wireless connectivity, and embedded vision for industrial control have contributed to the growing demand in the semiconductor industry. There is growing demand for mobile bandwidth, FPGA data centre accelerators, and vehicle electrification.

The integration of AI into EDA tools, global 5G rollout, and legislative actions like the CHIPS and Science Act are also shaping the market. Although there is an x86 and ARM duopoly in the global semiconductor market, at this growth rate, RISC-V will indeed balance the market.

As the traditional proprietary chips fell short and the industry hit its worst during Covid, RISC-V came to the rescue. The heightened demand and open-minded innovators led to the increased adoption of RISC-V ISA. In a way, the risk they took led to this victory and the coming. **EFY**

This article is compiled by Shivangi Kharoo based on the RISC-V Market Report by SHD Group.



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